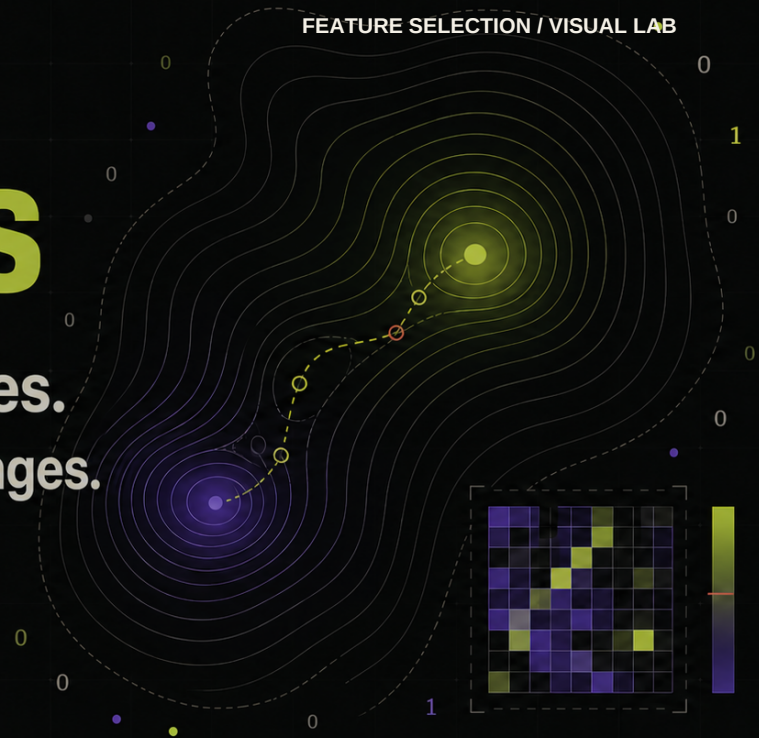


QUBOLens

Find the strongest features.
Drop the noise. See what changes.

A visual feature selection lab



THE PROJECT

A clearer way to choose which features deserve to stay.

QUBOLens is an open-source feature selection lab. Choose how many inputs to keep, then compare a smaller, less repetitive set with simple alternatives. Every result is visual, downloadable, and reproducible.

2

sample datasets

40

feature limit

5x

score comparison

WHY THIS IS USEFUL

Every extra input has a cost. Features can require sensor reads, database joins, API calls, memory, and time. A simple ranking can keep two features that tell nearly the same story.

QUBOLens looks at features together. It keeps features that connect with the target, avoids repeated information, and always respects the limit you choose.

PREMISE

My takeaway: quantum ideas can be useful before quantum hardware is. The practical first step is framing a messy choice as a clear optimization problem.

Useful now. Portable later.

selection = useful signal + less repetition + your limit

keep what helps / avoid what repeats / respect the feature limit

PIPELINE

01 Measure what helps

Each feature is checked against the target. Features are also compared with one another to find repeated information.

02 Try many combinations

A repeatable search explores different feature sets and gradually settles on the strongest option for the chosen limit.

03 Compare fairly

The chosen set is checked against a simple ranking and the full dataset, using the same score and data splits.

WHAT THE USER GETS

- An exact feature limit and adjustable overlap control
- A score-versus-size chart and live search trace
- Chosen features and an interaction map
- A fair comparison with simple alternatives
- A downloadable report and technical matrix
- Web UI, Python API, CLI, tests, CI, and Render deploy

CREDIBILITY CHECKS

The math is tested state-by-state. Seeds control the data, search, and splits. Every method uses the same comparison. The interface clearly labels scores as exploratory.

BOUNDARIES ARE PART OF THE PRODUCT

The included search is classical, not quantum computation. Correlation misses some nonlinear interactions. Comparison scores are not a held-out production estimate. A lower search score does not guarantee a better downstream model. QUBOLens says all four plainly.

RESEARCH BASIS

Mucke et al. (2023)

Feature Selection on Quantum Computers - Quantum Machine Intelligence.

Glover, Kochenberger & Du (2018)

A Tutorial on Formulating and Using QUBO Models.

Pranjic, Mummaneni & Tutschku (2024)

Quantum Annealing based Feature Selection in Machine Learning.

Hellstern, Dehn & Zaefferer (2023)

Quantum computer based Feature Selection in Machine Learning.

OPEN SOURCE / MIT

github.com/RS-010806/quantum

zero dependencies / one service